

NIAGARA FALLS INTL, NY, USA

WMO: **725287**Lat: **43.108N**Lon: **78.938W**Elev: **585**StdP: **14.39**Time Zone: **-5.00 (NAE)**Period: **94-19**WBAN: **04724**

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.9	6.5	-5.7	4.1	3.3	-1.1	5.3	7.8	32.6	29.0	29.0	27.8	10.3	260	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.2	88.0	72.2	85.4	71.0	82.6	69.6	75.2	83.7	73.5	81.6	72.0	79.7	13.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.4	122.7	79.7	70.9	116.4	78.3	69.7	111.4	77.1	38.9	83.7	37.4	81.7	36.1	79.6	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.7	23.9	20.6	DB	-3.8	91.4	6.1	2.7	-8.2	93.4	-11.8	95.0	-15.2	96.6	-19.7	98.6	(4)
(5)			WB	-4.4	77.9	6.0	2.2	-8.7	79.5	-12.2	80.8	-15.6	82.0	-19.9	83.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	24.6	25.9	33.6	45.4	57.5	67.3	71.8	70.2	63.1	51.6	40.3	31.1	(6)	
	DBStd	18.70	11.29	10.39	10.08	8.72	8.18	6.64	5.39	5.28	7.10	8.07	8.45	9.20	(7)	
	HDD50	3168	790	675	517	188	22	0	0	0	2	78	309	587	(8)	
	HDD65	6572	1254	1094	973	588	260	51	7	14	118	421	742	1050	(9)	
	CDD50	2681	1	1	9	52	255	518	677	625	395	128	18	2	(10)	
	CDD65	608	0	0	0	1	28	119	219	174	61	6	0	0	(11)	
	CDH74	4792	0	0	2	28	299	949	1757	1265	459	33	0	0	(12)	
	CDH80	1250	0	0	0	3	66	258	506	309	106	2	0	0	(13)	
(14) Wind	WSAvg	9.9	12.4	11.8	10.9	10.6	9.2	8.6	8.4	7.7	8.0	9.6	10.8	11.5	(14)	
(15) Precipitation	PrecAvg	35.3	2.7	2.1	2.6	3.0	3.1	2.8	3.4	3.2	3.3	3.0	3.1	3.1	(15)	
	PrecMax	43.0	4.7	4.6	4.8	5.2	5.5	6.2	7.3	5.4	7.0	7.4	4.7	6.6	(16)	
	PrecMin	28.0	1.4	0.9	1.2	1.3	0.7	0.7	1.0	1.5	1.8	0.9	1.1	1.3	(17)	
	PrecStd	4.1	1.0	0.9	0.9	1.2	1.4	1.5	1.6	1.2	1.3	1.5	0.9	1.2	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.5	58.1	69.7	78.7	85.6	90.0	91.3	90.4	88.0	78.1	66.4	59.7	(19)	
		MCWB	53.7	50.7	57.9	62.6	68.9	73.4	73.4	73.5	71.4	65.6	57.2	54.3	(20)	
	2%	DB	51.0	50.3	60.3	71.6	81.4	86.2	88.2	86.2	82.9	73.3	62.7	52.3	(21)	
		MCWB	48.1	44.9	51.1	57.5	66.6	70.9	72.8	72.0	70.2	62.8	54.9	48.0	(22)	
	5%	DB	45.1	44.9	54.5	65.6	77.2	82.7	85.4	83.3	79.3	69.4	58.6	48.1	(23)	
		MCWB	42.1	40.3	48.2	54.3	63.7	69.5	71.1	70.2	67.7	61.5	52.1	44.8	(24)	
	10%	DB	39.5	39.3	49.2	60.8	72.6	79.6	82.4	81.0	75.5	65.1	54.3	43.5	(25)	
		MCWB	36.3	35.8	43.4	51.3	60.7	67.1	69.6	69.3	66.0	58.5	48.9	39.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.1	53.0	59.1	64.1	71.3	76.3	77.9	77.4	74.3	68.5	59.4	53.9	(27)	
		MCDB	57.7	57.1	67.9	75.6	83.3	85.9	87.1	84.8	83.7	74.7	64.5	58.0	(28)	
	2%	WB	48.2	45.7	53.0	59.4	68.4	73.3	75.4	74.6	71.9	65.0	56.2	48.6	(29)	
		MCDB	50.4	49.7	58.2	68.5	79.1	83.0	84.2	82.1	79.8	71.1	61.2	51.5	(30)	
	5%	WB	42.1	40.4	48.6	55.9	65.7	71.3	73.7	73.0	69.9	61.8	52.9	44.9	(31)	
		MCDB	44.6	44.3	54.4	63.6	75.0	79.6	81.8	80.0	76.1	68.2	57.6	48.0	(32)	
	10%	WB	37.0	36.3	43.6	52.0	62.7	69.3	72.0	71.4	67.9	59.0	49.4	40.4	(33)	
		MCDB	39.4	39.5	48.5	59.4	70.0	76.8	79.5	78.2	74.2	64.4	54.3	43.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.9	14.0	15.4	18.8	20.2	18.7	18.2	18.0	18.6	16.2	14.0	11.4	(35)	
		MCDBR	17.1	20.6	23.8	27.4	25.0	21.6	20.4	20.4	21.9	21.9	20.6	16.3	(36)	
		MCWBR	15.6	16.9	16.7	17.0	13.4	11.0	9.6	9.8	11.4	13.2	15.3	13.8	(37)	
	5% WB	MCDBR	16.7	19.6	22.2	24.3	22.9	18.7	17.3	17.1	18.5	18.8	18.5	15.9	(38)	
		MCWBR	15.7	16.9	16.6	17.4	13.8	11.1	9.7	9.4	11.1	12.9	15.1	14.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.296	0.322	0.333	0.374	0.404	0.433	0.432	0.420	0.380	0.361	0.314	0.298	(40)	
	taud		2.417	2.340	2.407	2.332	2.290	2.240	2.260	2.282	2.375	2.410	2.507	2.472	(41)	
	Ebn at Noon		267	276	288	283	276	267	266	265	269	259	259	256	(42)	
	Edh at Noon		24	31	33	38	41	43	42	40	34	29	22	21	(43)	
(44) All-Sky Solar Radiation	RadAvg		416	666	1048	1401	1720	1853	1867	1656	1319	796	494	338	(44)	
	RadStd		39	61	111	149	181	140	143	88	117	74	68	43	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	-0.92	N/A	N/A	N/A	N/A	
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page